AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

M110
Python Programming

Midterm Examination (MTA)
Spring Semester 2024/2025

Date: 09/Apr./2025

Number of Exam Pages: 4 (including this cover sheet)	**Time Allowed:** 2 Hours

Instructions:

- Total Marks: **30**
- This exam consists of **2** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

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PART 1: Multiple Choices questions [5 Marks]

Question1: Answer the following questions by choosing the most correct statement.

1- What is the correct syntax to remove the element 23 from a list called nlist?
   a. remove.nlist[23]
   b. remove.nlist(23)
   c. nlist.remove[23]
   d. nlist.remove(23)

2- Among the choices below, which Python data type represents an ordered, mutable sequence of elements?
   a. Tuple
   b. List
   c. Float
   d. String

3- What is the output of ("aaa" < "aab")?
   a. error
   b. none
   c. True
   d. False

4- Which part of the expression **(x ** 2 + 6 / 3 *2)** is evaluated first?
   a. 3*2
   b. 6/3
   c. x**2
   d. 2+6

5- Which of the following is not a Boolean expression?
   a. 2.5<55
   b. x=24*2
   c. "x">="z"
   d. Both b and c.

PART 2: Problem Solving Questions [25 Marks]

Question 2: [10 marks]

What is the output of the following programs?

a- (2 marks)

```
n = 8
while(n>=0):
    if n%2==0:
        print(n, end=" ")
    n = n-2
print ("\nDone!")
```

b- (3 marks)

```
i=1
j=0
for i in range(2, 5):
    for j in range(1, i + 1):
        print('*', end='')
    print()
```


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c- (5 marks)

```
myString = "M110_Assessment"
myList=[3,7,5,8,4,2,1,6]
print(myString[:6])
print(myList[::2])
print(myString[-4:])
myList.insert(3,0)
print(myList)
myList.sort()
myList.reverse()
print(myList)
```

Question 3: [15 marks]

a- Write a Python program that prompts the user to enter a crochet hook size (B-G). The program should determine the corresponding metric diameter in millimeters using chained if-elif-else statements. (Assume that the entered value is an uppercase).

- If the input is valid (within the range B-G, where B, C, D, E, F, and G are valid sizes), the program should assign the proper value to the diameter.
- If the input is invalid (not within the range B-G), the program should assign the value -1 to the diameter.
- At the end of the program, you should print the metric diameter in millimeters if the diameter is not equal to -1, otherwise display an appropriate error message.

Size conversions for size: mm

- B : 2.25
- C : 2.75
- D : 3.25
- E : 3.5
- F : 3.75
- G : 4.0

Example:

- If the input is 'D', the program should print: "The diameter of a size D hook is 3.25 mm."
- If the input is 'A', the program should print: "Invalid hook size."

b- Write a Python program that performs the following operations on the numeric list: **numbers = [10, 20, 30, 40, 50]**

1. **Iteration (1 mark):** Iterate through the list and print all elements on one line.
2. **Search for the First Occurrence (2 marks):** Prompt the user to input a number and search for its first occurrence in the list. If found, print its index; otherwise, inform the user that the number is not on the list.

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3. **Remove a Specific Number (2 marks):** Prompt the user to input a number to remove from the list. If the number exists, remove its first occurrence; otherwise, inform the user that the number is not on the list.
4. **Insert a Number (2 marks):** Prompt the user to input a number and the position (index) where they want to insert it. Insert the number at the specified position. You must check whether the index position is valid or not before inserting the number.
5. **Displaying the Final list(1 mark)**: After performing all the operations, print the final state of the list.

Sample Output:

```
10 20 30 40 50
Enter a number to search: 5
The number 5 is not on the list.
Enter a number to remove: 20
The number 20 has been removed.
Enter a number to insert: 60
Enter the index position to insert the number: 6
Invalid index position.
[10, 30, 40, 50]
```

End of questions

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